An Analysis and Optimization of Flavin Mononucleotide Reactions and Structures as Pertaining to Thyroid Synthesis

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Abstract

Iodotyrosine deiodinase (IYD), a member of the nitroreductase superfamily, catalyzes the reductive dehalogenation of halotyrosines through two sequential single-electron transfers mediated by its flavin mononucleotide (FMN) cofactor. The catalytic specificity of this reaction depends critically on direct polar coordination between the substrate zwitterion and the isoalloxazine ring of FMN, which stabilizes the one-electron reduced flavin semiquinone (FMNsq) intermediate essential for halide elimination. Despite significant experimental characterization of these interactions, the structural geometry of the active site at the transition state and the electronic properties of the FMN cofactor along the reaction coordinate remain incompletely understood. Here, we employ molecular dynamics simulation and density functional theory calculations to investigate these properties computationally. MD simulations of oxidized IYD initiated from the crystal structure of Haliscomenobacter hydrossis IYD (PDB 5KO8) were performed over a 1 ns production trajectory using AmberTools, yielding a well-equilibrated system with key active site distances consistent with the polar coordination network required for FMNsq stabilization. DFT geometry optimizations of a truncated FMN hydroquinone system using the PBE exchange-correlation functional and a plane-wave basis set demonstrated convergence across the 600–700 eV cutoff range, with electron density mapping revealing preferential accumulation of negative charge at the carbon-carbon bond adjacent to the isoalloxazine nitrogen atoms, consistent with the theorized radical transition state. Together, these results provide computational support for the sequential single-electron transfer mechanism of IYD and establish a foundation for future transition-state modeling and in silico enzyme design.

Introduction and Background

The Nitroreductuse Superfamily

Flavin mononucleotide (FMN)-dependent enzymes represent one of the most catalytically versatile classes of biological catalysts known. Among these, the nitroreductase (NTR) superfamily a large group of enzymes that share a structural fold and a common dependence on FMN as their redox cofactor. Despite this structural conservation, members of the NTR superfamily catalyze a remarkable breadth of chemical transformations, including nitroaromatic reduction, enone reduction, quinone reduction, fatty acid reduction, flavin fragmentation, and reductive dehalogenation Copp et al., 2018. This functional diversity, arising from a single conserved architecture, makes the NTR superfamily a particularly powerful model system for studying how subtle changes in protein environment can redirect cofactor chemistry toward entirely different chemical outcomes.

A unifying feature of all NTR superfamily members is their use of a ping-pong bi-bi kinetic mechanism, in which the FMN cofactor is first reduced by an electron donor substrate and subsequently reoxidized by an electron acceptor substrate in separate half-reactions Sun et al., 2017. The cofactor therefore cycles between its oxidized and reduced forms across each full catalytic turnover. The specific chemistry executed during each half-reaction whether a single electron is transferred stepwise or two electrons are transferred simultaneously as a hydride depends critically on the electronic environment established around the FMN isoalloxazine ring system by the surrounding protein and its bound substrates Mukherjee and Rokita, 2015; Lemen and Rokita, 2024. Understanding how individual NTR superfamily members achieve their

distinct functional identities from a common chemical scaffold is therefore a central problem in enzymology and has broad implications for enzyme engineering and protein design.

Within the NTR superfamily, the iodotyrosine deiodinase (IYD) subgroup is distinguished by its ability to catalyze reductive dehalogenation , more specifically, the removal of a halogen substituent from an aromatic ring and its replacement with a hydrogen atom Sun et al., 2017; Mukherjee and Rokita, 2015,. In vertebrates, IYD performs this reaction on mono- and diiodotyrosine (I-Tyr and I_2-Tyr), which are generated as byproducts during the biosynthesis of the thyroid hormones thyroxine (T4) and triiodothyronine (T3). Because iodide is scarce in most terrestrial environments and essential for thyroid hormone production, IYD provides a critical iodide recycling function: by cleaving the carbon-iodine bond of iodotyrosines, the enzyme recovers free iodide that can be reincorporated into new thyroid hormone. Genetic deficiency of IYD results in hypothyroidism, and mutations in the human IYD gene have been identified in patients with goiter and developmental defects Sun et al., 2017.

Beyond its role in thyroid hormone metabolism, IYD has attracted significant interest for its potential application in environmental bioremediation. Halogenated aromatic compounds, including chlorinated and brominated phenols and tyrosines, are widespread environmental contaminants arising from industrial processes and immune-mediated halogenation in tissues Sun et al., 2017; Bobyk et al., 2015. IYD has been shown to catalyze the dehalogenation of not only its canonical iodotyrosine substrates, but also bromotyrosine and chlorotyrosine, demonstrating a broader substrate scope than originally recognized Bobyk et al., 2015. This capacity for reductive dehalogenation of environmentally relevant compounds positions IYD as a promising scaffold for the development of enzymatic tools for bioremediation, provided that its substrate scope and catalytic efficiency can be further optimized Sun et al., 2017.

The Mechanism of Reductive Dehalogenation

The chemical mechanism by which IYD accomplishes reductive dehalogenation is fundamentally dependent on the FMN cofactor, which serves as the primary electron donor to the halogenated substrate Sun et al., 2017; Bobyk et al., 2015. The overall reaction proceeds through two sequential single-electron transfer steps, distinguishing IYD from most other dehalogenases, which operate via hydrolytic or oxidative mechanisms and from anaerobic reductive dehalogenases that depend on cobalamin or other metal-containing cofactors Bobyk et al., 2015.

In the accepted mechanism, the halotyrosine substrate binds to IYD in its phenolate form, guided by direct coordination of the substrate's zwitterionic -ammonium and carboxylate groups to specific positions on the flavin isoalloxazine ring Lemen and Rokita, 2024. Once the enzyme-substrate complex is formed and the active site lid has closed over both the substrate and the flavin, the reduced form of FMN (FMNhq) donates a single electron to the substrate. This generates the one-electron-reduced flavin semiquinone (FMNsq) and a transient substrate radical intermediate Sun et al., 2017; Mukherjee and Rokita, 2015, . This substrate radical is inherently unstable and rapidly undergoes reductive cleavage of the carbon-halogen bond, releasing the halide ion as a leaving group and generating a product radical. A second electron transfer from FMNsq to the product radical then restores aromaticity and yields tyrosine as the dehalogenated product, while FMN is returned to its fully oxidized state (FMNox), completing the catalytic cycle Sun et al., 2017; Bobyk et al., 2015.

The sequential single-electron transfer mechanism has been supported by multiple lines of experimental evidence. Replacement of the native FMN cofactor with 5-deazaFMN, an analog structurally identical to FMN but incapable of single-electron transfer chemistry, reduced catalytic efficiency by more than four orders of magnitude, demonstrating the strict dependence of dehalogenation on one-electron chemistry Lemen and Rokita, 2024. Rapid kinetic studies further revealed that the rate of flavin oxidation by bromo- and iodotyrosine is similar despite the differing strengths of their carbon-halogen bonds, suggesting that the chemical step of halide elimination does not limit catalysis for these substrates and that initial electron transfer or a pre-

ceding proton transfer step may instead determine the rate. Chlorotyrosine, which has the strongest carbon-halogen bond among active substrates, reacted approximately 20 times more slowly and generated a transient spectral intermediate, consistent with a shift in the rate-limiting step toward halide elimination as the bond strength increases Bobyk et al., 2015.

The Redox States of FMN and Their Functional Importance

The ability of FMN to support the sequential single-electron transfer mechanism of IYD rests on its unique capacity to exist stably in three distinct oxidation states, each with characteristic spectroscopic properties that allow monitoring by UV-visible absorption and fluorescence spectroscopy Sun et al., 2017; Mukherjee and Rokita, 2015. The fully oxidized form, FMNox, absorbs light strongly at approximately 450 nm and displays characteristic fluorescence emission at 516 nm when excited at 450 nm Lemen and Rokita, 2024. This fluorescence is substantially quenched upon substrate binding to the IYD active site, providing a convenient spectroscopic handle for monitoring binding kinetics and affinity Lemen and Rokita, 2024. The fully reduced, two-electron hydroquinone form, FMNhq, lacks absorbance in the visible region beyond 400 nm and is the catalytically active species that initiates dehalogenation by donating the first electron to the substrate Sun et al., 2017.

The one-electron-reduced neutral semiquinone, FMNsq, is the key intermediate in IYD's catalytic cycle. It is spectroscopically distinguishable from both FMNox and FMNhq by a broad absorbance extending to approximately 600 nm that is diagnostic of the neutral radical form Sun et al., 2017; Mukherjee and Rokita, 2015. The accumulation of FMNsq during reductive titration experiments — conducted by progressively reducing the enzyme with a slow, controlled reductant such as the xanthine/xanthine oxidase system — serves as an experimental readout of whether the enzymatic environment thermodynamically stabilizes the semiquinone state Lemen and Rokita, 2024. In the absence of any substrate or ligand, IYD fails to accumulate detectable FMNsq during reductive titration, and the enzyme's midpoint potential closely resembles that of free FMN in solution Sun et al., 2017. This observation indicates that the protein scaffold alone is insufficient to stabilize the semiquinone and that an additional factor — the substrate — is required to shift the redox thermodynamics in a way that favors the radical intermediate essential for dehalogenation Sun et al., 2017; Lemen and Rokita, 2024.

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Polar Coordination as the Necessary Condition for Semiquinone Stabilization

The substrate-dependent stabilization of FMNsq in IYD is mediated through a network of polar contacts that form between the isoalloxazine ring of FMN and specific functional groups present in both the protein and the bound substrate. Crystal structures of IYD in complex with I-Tyr reveal three principal polar interactions with the flavin: a hydrogen bond donated from the side chain of a conserved threonine residue to the N5 position of the isoalloxazine, coordination of the substrate's -ammonium group to the C4 carbonyl oxygen (O4) of the flavin, and coordination of the substrate's carboxylate group to the N3 position of the flavin. These three interactions collectively remodel the electronic environment of the isoalloxazine ring in a way that raises the one-electron midpoint potential of FMN and thermodynamically favors the semiquinone state Sun et al., 2017.

Initial investigations focused on the threonine-to-N5 hydrogen bond as the primary mechanistic determinant, following a well-established paradigm in flavoenzymology where hydrogen bonding to the N5 position is the dominant switch controlling one- versus two-electron transfer chemistry Mukherjee and Rokita, 2015. Consistent with this model, substitution of the conserved Thr with Ala in *H. hydrossis* IYD eliminated FMNsq accumulation and shifted the enzyme's activity from dehalogenation to nitroreduction, the reaction characteristic of NTR superfamily members that use two-electron hydride transfer Mukherjee and Rokita, 2015. However, the decrease in dehalogenation efficiency caused by this Thr-to-Ala mutation was only 15-fold, substantially less than the catastrophic loss caused by 5-deazaFMN substitution Lemen and Rokita, 2024. This disparity indicated that the threonine-N5 interaction is not the sole or even primary determinant of single-electron chemistry in IYD, and that the substrate's own contribution to the polar environment of the flavin was being underestimated.

Substrate Zwitterion as the Dominant Controller of Flavin Chemistry

The critical role of the substrate zwitterion in controlling flavin chemistry was established by systematic modification of the -ammonium and carboxylate groups of I-Tyr in studies using bacterial IYD Lemen and Rokita, 2024. Neutralization or deletion of the carboxylate group had negligible effects on binding affinity for the resting FMNox enzyme, but suppressed catalytic efficiency for dehalogenation by over 2,000-fold and eliminated FMNsq accumulation during reductive titration, demonstrating that carboxylate coordination to the flavin N3 is essential for stabilizing the semiquinone during catalysis even though it contributes little to ground-state binding. Removal of the -ammonium group abolished detectable binding entirely, establishing this cationic group as the primary anchor for substrate recognition through its interaction with the flavin O4 Lemen and Rokita, 2024.

Critically, the same substrate modifications that suppressed dehalogenation and FMNsq formation concurrently activated a latent nitroreductase activity in wild-type IYD. Substrates lacking the full zwitterionic structure were capable of fully reducing the nitro group of nitrityrosine derivatives to a hydroxylamine product via hydride transfer. which is the two-electron mechanism characteristic of conventional nitroreductases, while the same substrates in their intact zwitterionic form were efficiently dehalogenated by single-electron transfer. This reciprocal switch in chemoselectivity, paralleling the effect of the Thr-to-Ala protein mutation, demonstrates that the substrate itself functions as the primary configurator of the flavin's electronic state and dictates which of its latent chemical functions is activated. IYD is therefore unusual among flavoproteins in placing the responsibility for cofactor activation largely on the substrate rather than the protein, a design that provides an elegant mechanism for ensuring that only free halotyrosines that was not embedded in peptides or proteins are processed by the enzyme Lemen and Rokita, 2024.

These findings establish IYD as a compelling model system for investigating how polar interactions at the flavin isoalloxazine ring modulate re-

dox chemistry and electron transfer specificity. A complete mechanistic understanding, however, requires knowledge not only of which interactions are present in the ground-state enzyme-substrate complex, but also of how the electronic structure of the flavin evolves along the reaction coordinate and what the geometry of the active site looks like at the transition state. Computational approaches, specifically density functional theory (DFT) calculations to analyze the electron density of FMN in its semiquinone state and molecular dynamics (MD) simulations to sample the geometry of the catalytic environment at the transition state offer a powerful complement to the experimental investigations described above, and constitute the focus of the present work.

Molecular Dynamics Simulation

We performed MD simulations of Iodotyrosine Deiodinase in its oxidized form within a water solvent. Here, we focus on the active site geometry configuration at the transition state, which was experimentally investigated by Rokita et al. [1] The configuration of the active site in the transition state is designed to mimic the key bonds and interactions between the FMN cofactor, side chain groups of specific residues, and the zwitterion of iodotyrosine (substrate).

To further characterize the active-site arrangement associated with the stabilization of the intermediate semiquinone, the crystal structure of oxidized iodotyrosine deiodinase was extracted from PDB 5KO8 and alnalyzed in the context of previously reported studies of the flavin semiquinone intermediate. Based on literature describing the transition-state stabilization, several key hydrogen bondning and electrostatic interactions involving the FMN cofactor, active-site residues, and the iodotyrosine zwitterion were identified and used as reference geometries for the MD simulation (Figure 1).

The carboxylate and ammonium groups of substrate zwitterion and the N5/O4 positions of the flavin ring, the guanidine side chain group of arginine and the N1/O2 position of the flavin ring, the side chain group of threonine and the N5 position of the flavin ring and lastly the pi-stacking of the phenolic ring of the substrate over the isoalloxazine

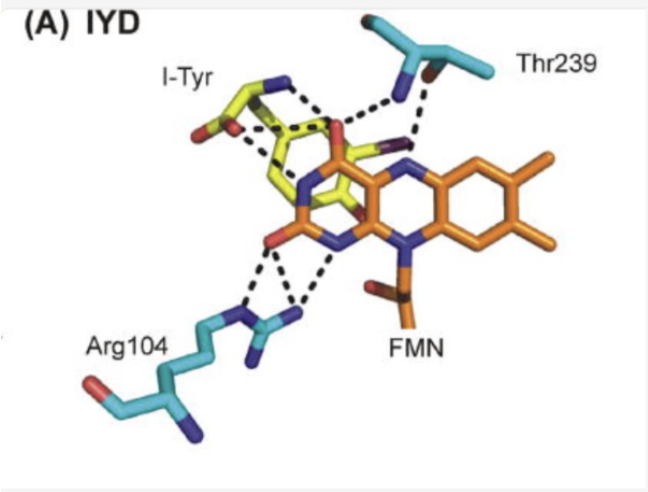


Figure 1. Active site configuration of Iodotyrosine Deiodinase (IYD) with bound iodotyrosine (I-Tyr) substrate. The FMN cofactor (orange) is shown in a stacking interaction with the I-Tyr substrate (yellow), with dashed lines indicating key hydrogen bonding and electrostatic interactions between the substrate zwitterion and active site residues. Arg104 (cyan) forms contacts with the carboxylate group of the I-Tyr zwitterion, while Thr239 (cyan) interacts with the amino group, together orienting the substrate for productive catalysis. All interactions are depicted as black dashed lines.

ring of FMN were all the interactions necessary for semi quinone stabilization.

Our molecular dynamics simulations were performed using AmberTools simulation package. System preparation, parameterization, solvation, and trajectory generation were carried out using AmberTools workflows for biomolceular simulations.

The starting structure was the oxidized iodotyrosine deiodinase which was prepared from the crystallographic coordinates of PDB 5KO8.I nitial system preparation involved removal of crystallographic water molecules present in the PDB structure to minimize potential steric clashes during setup and equilibriation. Next, the complex was separated into its primary components , which were the FMN cofactor, iodotyrosine substrate, and the protein backbone containing all amino acid residues. Following system parsing, missing hydrogen atoms were added to the structures prior to

parameterization and topolgy generation.

Although AmberTools provides established force-field parameters for standard protein amino acid residues, additional parameterization steps were required for non-standard molecular components present in the system, specifically the FMN co-factor and the iodotyrosine substrate. These ligands were treated separately using external charge derivation and force-field parameterization workflows to generate partial charge parameters. Partial atomic charges for the FMN and IYR structures were assigned using the AM1-BCC semi-empirical charge method, after which missing force-field parameters were generated and incorporated into the complete simulation topology prior to solvation and equilibration.

Following parameterization, the complete protein-ligand complex was solvated using a water box model within AmberTools (Figure 2). The parameterized system was placed in a periodic solvent buffer extending 10 Angstroms in all directions. The solvent was generated to provide an aqueous environment for the simulation. Counter ions consisting of sodium and chloride ions were added using AmberTools ions placement routines to neutralize the net charge of the system and stabilize long-range electrostatic interactions under periodic boundary conditions.. The fully solvated and neutralized structure was then used as the starting configuration for the energy minimization, heating, equilibration, and production MD simulations.

Following solvation and ion neutralization, the system was heated gradually from 0 K to 300 K over a 100 ps simulation period using AmberTools molecular dynamics engine. The heating statge was performed using 50,000 simulation time steps with a 2 fs step size. Initial coordinate information from the solvated structure was used as the starting configuration for the simulation, while velocities were recording during the heating process. Periodic boundary conditions were applied under a constant volume (NVT) ensemble conditions with an 8.0 Angstrom non-bonded interaction cutoff. Temperature regulation was maintained using a Langevin thermostat with a collision frequency parameter of 2.0 $ps^{-}1$ to ensure stable thermal equilibration of the solvated system.

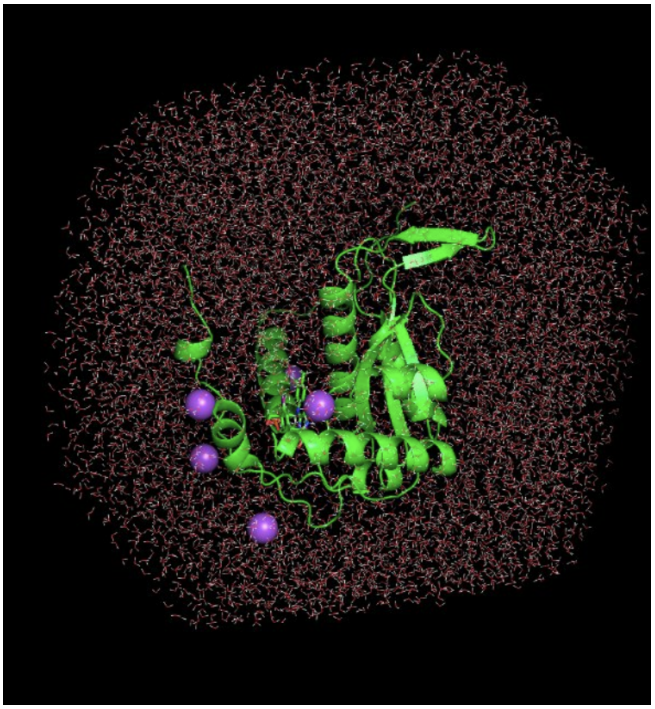


Figure 2. Solvation of the protein system prior to molecular dynamics simulation. The protein structure (green ribbon) is shown enclosed within a cubic water box (10), with explicit TIP3P water molecules represented as red/white sticks. Purple spheres indicate sodium (Na^{+}) ions added to neutralize the net charge of the system. The solvated system was prepared using AmberTools to ensure periodic boundary conditions with sufficient solvent buffering around the protein.

Following the heating stage, the system was equilibrated under constant-pressure (NPT) and constant-temperature (NVT) conditions using the final coordinates obtained from the heating simulation as the starting structure. Equilibration was performed for 100 ps using 50,000 simulation steps with a 2 fs integration time step within AmberTools environment. Pressure was maintained using the Berendsen baraostat with a pressure relaxation time constant of 2.0 ps, while the temperature control at 300 K was achieved using a Langevin thermostat. Bond lenths involving hydrogen atoms were constrained using the SHAKE algorithm, allowing stable integration at the desired time step throughout the equilibration procedure.

Following equilibriation, a 1 ns production molecular dynamics simulation was performed using the equilibriated structure and velocities as the starting conditions within the AmberTools simulation environment. The production phase consitisted of 500,000 simulation stepes with a 2 fs integration timstep, correspionding ro a toal simulation time of 1 ns. Simulations were conducted under constant-volume and constant-temperature (NVT) condiitons using an 8.0 Angstrom nonbonded interaction cutoff and a Langevin Thermostat to maintain the target temperature at 300K. Bond lenths involving hydrogen atoms were constrained using the SHAKE algorithm throughout the simulation. The system energies and trajectory coordinates were recorded every 10 ps, yielding a total of 100 trajectory frames used for structural and active-site analyses. Alongside trajectory generation, several thermodynamic observables were monitored during the production simulation to assess system stability and convergence. Averaged quantities including total energy, kinetic energy, and potential energy were recorded throughout the trajectory, together with additional computational metrics such as the Ewald error estimate for evaluation of long-range electrostatic accuracy under periodic boundary conditions. Alongside trajectory generation, several thermodynamic observables were monitored during the production simulation to assess system stability and convergence. Averaged quantities including total energy, kinetic energy, and potential energy were recorded throughout the trajectory, together with additional computational metrics such as the Ewald error estimate for evaluation of long-range electrostatic accuracy under periodic boundary conditions (Figure 3).

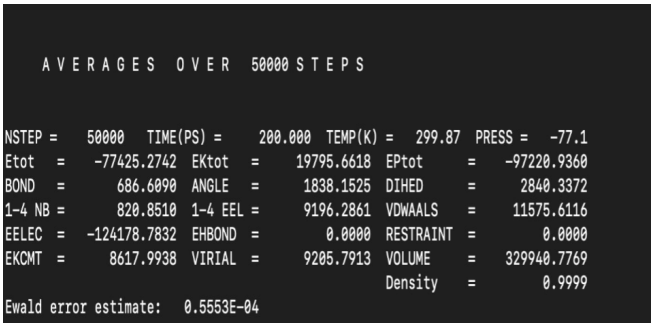


Figure 3. Summary statistics averaged over the NPT production MD simulation (50,000 steps, 100 ps). Thermodynamic and energetic averages are reported at the conclusion of the production run. The system reached a mean temperature of 299.87 K, approaching the target of 300 K, with a total energy (Etot) of 77,425.27 kcal/mol and a potential energy (EPtot) of 97,220.94 kcal/mol. The system density converged to 0.9999 g/cm^3, consistent with bulk water at 300 K and confirming proper equilibration under NPT conditions. The Ewald error estimate of 5.55 × 10 indicates sufficient accuracy in the long-range electrostatic calculations. Output generated using AmberTools

Trajectory frames generated during the production simulation were subsequently analyzed to identify energetically favorable active-site conformations based on potential energy values extracted from the MD trajectory.

Results

The trajectory frame related to the minimum potential energy observed during the production simulation was extracted for detailed structural analysis (Figure 4). This configuration was selected to represent low-energy active-site arrangement and was used to examine hydrogen-bonding interactions, substrate orientation and FMN coordination geometries relevant to the proposed transition-state stabilization mechanism. The analysis of the production trajectory showed a minimum potential-energy configuration at simulation step 225,000, corresponding to 650 ps of the 1 ns production run. The extracted structure showed a potential energy

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NSTEP =   225000   TIME(PS) =     650.000  TEMP(K) =   301.73  PRESS =     0.0
Etot   =    -78080.7056  EKtot   =     19918.3198  EPtot      =   -97999.0254
BOND   =       700.3980  ANGLE   =      1871.2736  DIHED      =     2773.5313
1-4 NB =       810.3632  1-4 EEL =      9166.5808  VDWAALS    =    11530.5655
EELEC  =   -124851.7378  EHBOND  =         0.0000  RESTRAINT  =        0.0000
Ewald error estimate:   0.1725E-03
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Figure 4. Instantaneous thermodynamic output at the lowest potential energy frame of the MD production run (Step 225,000; t = 650 ps). At this snapshot, the system recorded its minimum potential energy (EPtot = 97,999.03 kcal/mol) observed during the simulation, with a corresponding total energy (Etot) of 78,080.71 kcal/mol and kinetic energy (EKtot) of 19,918.32 kcal/mol.

of 97,999.0254 kcal/mol and was chosen as the representative low-energy active-site conformation for geometric and interaction analyses. The structure of this configuration was used to evaluate the spatial arrangement of the FMN cofactor, iodotyrosine substrate, and neighboring catalytic residues associated with transition-state stabilization.

The structure corresponding to the minimum potential energy sampled during the molecular dynamics trajectory was extracted and analyzed to evaluate active-site geometry and coordination interactions with the FMN flavin ring (Figure 5). The resulting configuration revealed several intermolecular distances between the flavin cofactor, iodotyrosine substrate, and neighboring catalytic residues that are consistent with proposed transition-state stabilization interactions. The guanidinium side chain of arginine was positioned proximal to the N1 and O2 atoms of the flavin ring, with interaction distances of 5.4, 6.1, and 6.5 Å, respectively. In addition, the substrate carboxylate group remained within hydrogen-bonding proximity of the N3 and O4 positions of FMN, exhibiting distances of 4.1 and 3.2 Å. The substrate ammonium group was located 2.9 Å from the O4 atom of the flavin ring, suggesting a strong electrostatic or hydrogen-bonding interaction. Furthermore, the hydroxyl-containing side chain of threonine exhibited a distance of 4.3 Å relative to the N5 position of FMN. Collectively, these geometric features support preservation of key active-site interactions implicated in stabilization of the flavin semiquinone transition-state configuration.

DFT Analysis

Computational Methods

Density Functional Theory Kohn-Sham Density Functional Theory (DFT) has become a major method of performing quantum chemical analyses in the modern age. Stemming from the original Hohenberg and Kohn DFT, Kohn and Sham were able to refine electron-electron interactions to isolate the exchange-correlation functional. This development allowed for more accurate approximations of electron density (via exchange-correlation energy), especially for larger systems, by avoiding the need for a direct solution to the Schrödinger wave function.

For this line of work DFT calculations were done utilizing the Atomic Simulation Environment (ASE) and the GPAW software. Additionally, the Perdew-Burke-Ernzerhof (PBE) functional was used to approximate the exchange-correlation.

Exchange-Correlation Functional The PBE exchange-correlation functional is a Generalized Gradient Approximation (GGA) level algorithm used for calculating electron density. GGA models improve upon local density approximation (LDA) models by adding in an additional term based on the gradient of electron density.

$$E_{xc} = E(\rho) \tag{1}$$

$$E_{xc} = E(\rho, \nabla\rho) \tag{2}$$

Shown above in Equation 1 is the general LDA model for electron exchange-correlation, and in Equation 2 that of a general GGA model. Because GGA models allow for the extra term, $\nabla\rho$, a more precise approximation of electron density is given by such. This precision is often important in organic catalysis as such molecules' electrons typically behave with significant nuance compared to a standard metallic catalyst.

The additional precision of a GGA model, however, is not without its expenses. GGA and higher level models (such as meta-GGA) have more terms used to approximate exchange-correlation, leading to computation times often spanning orders of magnitude with respect to their lower level counterparts. This is particularly pertinent to systems dealing with

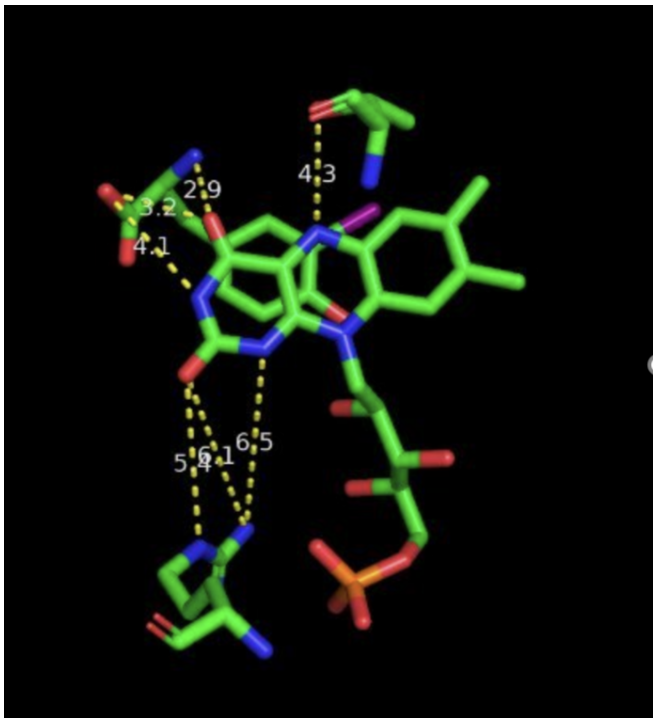


Figure 5. Active site geometry of the lowest potential energy snapshot (t = 650 ps) extracted from the MD production run, showing key atomic coordination distances (Å). The FMN cofactor (green/blue) is shown alongside the bound substrate and coordinating residues, with yellow dashed lines indicating interatomic distances measured in Angstroms. Short-range contacts of 2.9 and 3.2 Å suggest hydrogen bonding interactions at the substrate binding interface, while distances of 4.1, 4.3, 5.4, 6.1, and 6.5 Å reflect longer-range electrostatic or van der Waals coordination contacts. Visualization performed in VMD.

medium to large molecules, as these processes often do not scale linearly with respect to size of molecules, but at higher orders.

With the molecule of interest for this work being Flavin Mononucleotide (FMN) Hydroquinone (Hq), PBE was selected as a preliminary screening functional to be a balance between computational efficiency and accuracy. Because of the size of the FMN Hq system, a truncated subsection (T-FMN Hq) of the full molecule was chosen (seen in [figure reference]) for analysis. With a system of 34 atoms ($C_{14}H_{13}N_4O_3$) for T-FMN Hq, it was imperative to select an initial functional that could reliably solve systems of similar sizes.

Geometry Optimization In order to perform a geometry optimization in a medium-sized molecular system, a sufficient algorithm is needed to properly relax the geometry. For this project, the selected algorithm was the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm. This process utilizes Newton's method in higher dimensions, but importantly approximates the hessian matrix and its inverse to save computations. By calculating the forces experienced by bodies in the system, this process revolves around solving for the electron density to iterate through the gradual movement of atoms into their relaxed positions.

The factors for this analysis used to assess convergence is a combination of structural, electrical, and energy convergence elements. Solved geometries are compared to one another via HOMO/LUMO gaps, atom positions, and overall DFT energy to determine how plane-wave energy cutoff thresholds impact convergence in future calculations.

Functional Parameters While the exchange-correlation functional plays a large role in determining calculations, it is far from the only relevant parameter. An additional requirement for a complete calculation is a basis set representation. Because basis sets approximate electron wavefunctions differently, in order to analyze convergence, it becomes integral to choose a basis set representation that facilitates straightforward and intuitive precision of measurements. For that reason, a plane-waves basis set was chosen to be the representation and are given

by the summation of periodic functions to create a Fourier function. This choice is physically relevant as it provides a direct way to compare states of convergence via energy cutoff values. As opposed to another basis representation where convergence may be more ambiguous, plane-waves can be directly compared with different energy cutoff values to determine model convergence.

Structural Comparison Methods In order to assess structural similarity between molecules, first a universal method must be used to compare the differences in bond lengths. For this project, the method of doing so was to use the root mean square deviation (RMSD) to compare the geometries of two given molecules. This is a process which compares two given geometries of identical formulae (necessarily so) via the systematic comparison of the atomic positions of one molecule to its corresponding atomic positions in the other.

From this method, another difficulty arises in how to reliably and consistently compare two molecules with the same structure with differently optimized geometries. The solution to in this analysis is to use the Kabsch algorithm, a method of minimizing RMSD between to compared sets of points via orientation. This is important for our system as it preserves molecular geometry, and only allows for rotational and translational transformations.

Results and Discussion

Energy Cutoff Convergence As seen in Figure 1, as plane-waves cutoff energy increases for a given system, the basis tends towards completeness. Further seen in Figure 2, as the basis set tends towards completeness, the system reaches a point of diminishing returns. As seen in Figures 1 and 2, energies and successive energy differences show a clear path that suggests a converged state is being approached which is consistent with basis completeness.

HOMO-LUMO Gap Convergence Electrical convergence is an important and often overlooked factor for determining overall geometric convergence for a given system. In Figure 3, the HOMO-LUMO gap of the molecule can be seen to stay relatively consistent, with the caveat of the anomaly at the 650 eV plane-wave energy cutoff system.

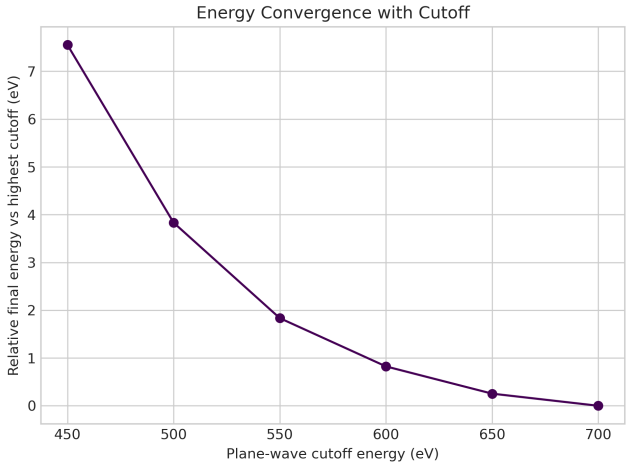


Figure 1: Relative final DFT energy of optimized T-FMN Hq structures as a function of plane-wave cutoff energy. Energies are referenced relative to the 700 eV calculation. Increasing cutoff energy produced diminishing improvements observed at approximately 600–650 eV, suggesting approach toward convergence.

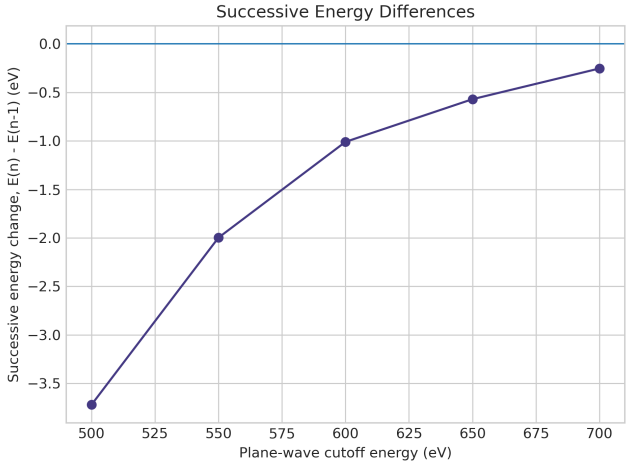


Figure 2: Sequential differences are seen in optimized geometries' total energy ($E_n - E_{n-1}$). Decreasing magnitude of successive plane-wave cutoff energies indicates approach on energetic convergence.

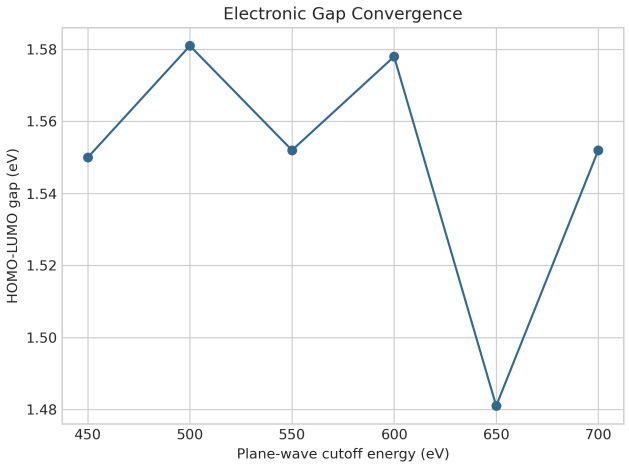


Figure 3: HOMO-LUMO gap of optimized T-FMN Hq structures as a function of plane-wave cutoff energy. While the gap remained relatively stable (1.565 ± 0.02eV), a deviation at 650 eV suggests some sensitivity in electronic structure to neighboring local minimums.

Structural Convergence Structural convergence is another often overlooked aspect of computational quantum chemistry that has very high relevance and often small discrepancies in geometry can result in large alterations of other structural properties. For this work, a root mean square deviation analysis was used to compare how geometries converge over time. As seen in Figure 4, RMSD appears to converge with a distinct anomaly at the 650 eV plane-wave cutoff energy geometry.

Force Convergence Force convergence is another metric important to the assessment of geometric convergence. If too many atoms are experiencing high forces, one is likely unable to conclude that they have indeed found a stable geometry. This makes claiming any properties about the ostensibly solved geometry significantly more difficult, and as a result detracts from the quality of calculations.

Computational Cost Scaling Plane-wave coefficients often correlate heavily with computation time for a system. This creates a trade off between higher cutoff energy parameter and producing physically meaningful results. As evident in Figure 6, a large number of steps can be required for each geometry solution. When compared with Figure

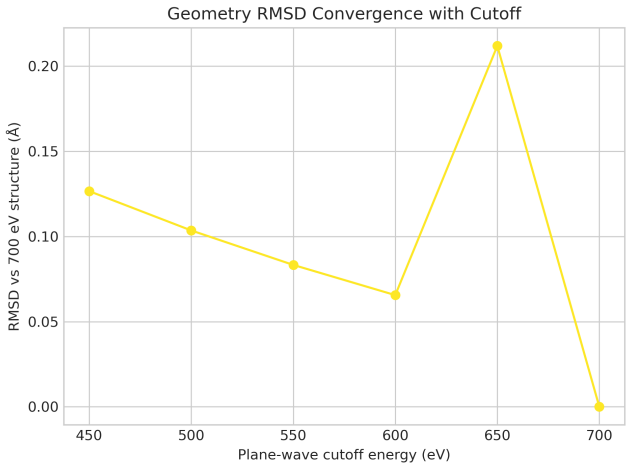


Figure 4: Root mean square deviation (RMSD) between successive optimized geometries and reference of structure optimized at 700 eV. Most tested cutoff energies display gradual structural convergence; however, the particularly large deviation at 650 eV strongly suggests a separate local minimum was identified.

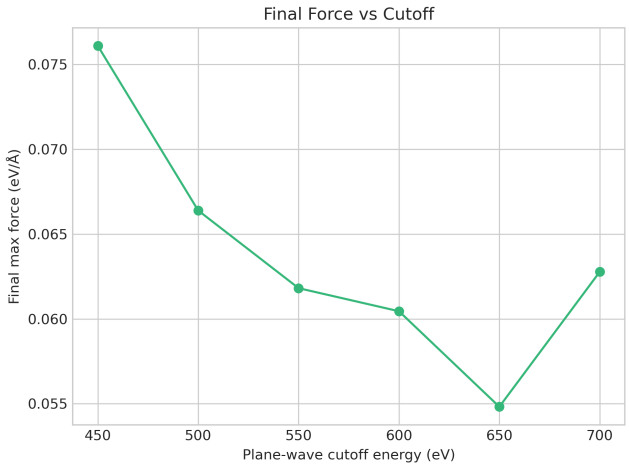


Figure 5: Maximum atomic force after geometry optimization as compared to plane-wave cutoff energy. Residual forces appear to decrease relative to increasing cutoff energy. A decrease at 650 eV suggests that the aforementioned hypothesized local minimum has a moderately more relaxed structure.

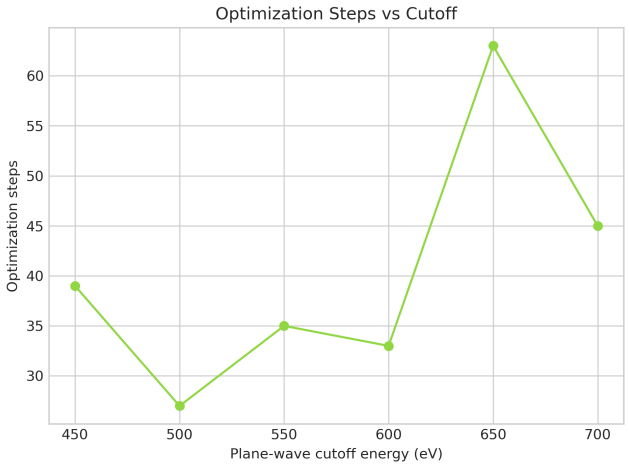


Figure 6: Number of BFGS optimization steps required for structural convergence at each given plane-wave cutoff energy. The increased optimization at 650 eV also suggests an alternative local minimum and by proxy further implies sensitivity of electronic structure convergence.

7, the number of plane-wave coefficients requires increases as well, often leading to higher computational expenses.

Electron Density Mapping To properly model the anionic molecule FMN Hq, the dft model should be able to identify in which areas the charge will accumulate and deplete from. Pictured in 8, it can be seen that negative charge has a tendency to accumulate around the carbon-carbon bond adjacent to the nitrogen atoms of the molecule. This is consistent with the theorized transition state that possess a free radical on said carbon complex.

Limitations

Basis Set and Cutoff Limitations Due to the nature of using plane-waves as the basis set, completeness is always approached in the infinite limit, and never truly reaches the theoretical ideal state. However, this does not disqualify the usefulness of such a model. Convergence is multifaceted and as so, convergence can be approached in different manners which are not necessarily dependent upon one another. In fact, as shown, the use of solely one metric to assess convergence can produce borderline misleading results.

If one were to use an energetic convergence met-

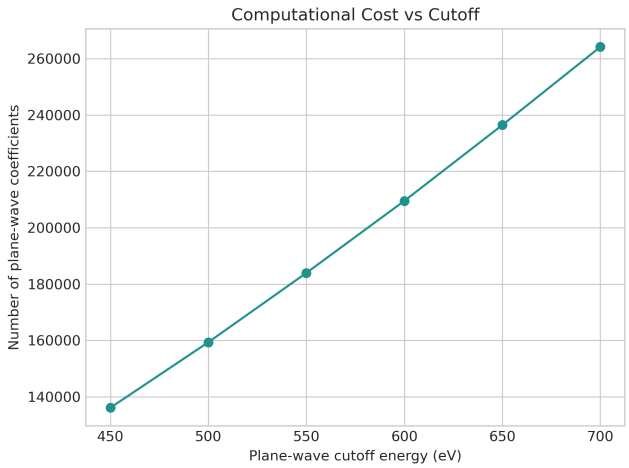


Figure 7: Number of plane-wave coefficients used in the basis representation relative to respective cutoff energies. Increasing cutoff energies substantially increased basis size which correlates with higher computational costs associated with solutions.



Figure 8: Front and back view of 700 eV plane-wave cutoff energy system optimized geometry of T-FMN Hq with red graphics indicating electron accumulation, and blue indicating electron depletion with respect to the difference between the charged state versus the neutral. The carbon-carbon bond adjacent to nitrogen atoms appears to show the greatest net accumulation of electron density.

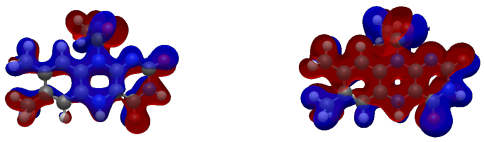


Figure 9: Electron density difference maps between Kabsch aligned optimized geometries. The electron density map of the 700 eV plane-wave cutoff energy system is compared to the density maps of the 600 eV (left) and 650 eV (right) plane-wave cutoff energy runs. Red and blue isosurfaces represent positive and negative density differences respectively. The broader distribution of the 650 eV calculation suggests greater deviation from the electronic structure of the 700 eV calculation.

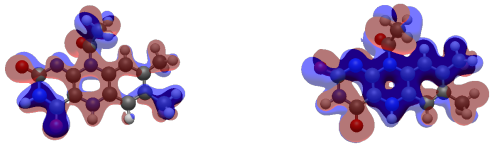


Figure 10: Alternate orientation of the electron density difference comparisons shown in Figure 9. This comparison continues to suggest that the 650 eV plane-wave cutoff energy system's electronic structure deviates more strongly from the 700 eV plane-wave cutoff energy system's than the 600 eV plane-wave cutoff energy system's does.

ric, there would be no discrepancies between the set of tested plane-wave cutoff energies. When electronic and and structural factors are included, convergence becomes more difficult (yet still possible) to assess.

Local Minimum Sensitivity One particularly interesting part of these studies was an anomalous data point for the 650 eV plane-wave cutoff energy system. As seen in both Figure 3 and in Figure 4, the overall solution exhibits behavior consistent with an alternative local minimum. This behavior could be relevant if it is shown to be physically relevant. If there is indeed a nearby local basin, this could directly impact future modeling as multiple distinct, yet similar structures could exist in tandem. It is also possible that only one structure exists in reality. If the former is so, the deviation in the HOMO-LUMO gap could be significant enough to impact material properties.

Future Directions

Other Relevant Molecules While T-FMN Hq is the most important molecule to understand for the given system, there exist many other relevant molecules. The two counterparts of FMN Hq, FMN Sq and FMN Ox, would allow for a deeper analysis on the overall oxidization pathways relevant in Thyroid syntheses. Difficulties in this sphere would likely arise from the free radical seen in the predicted transition state of the FMN co-factor. Because of this, it would be important to factor in spin polarization as this would no longer be a closed shell system.

In addition to the relevant molecules in the FMN family, I-Try, Thr239, and Arg104 are all necessary for the process of Thyroid synthesis, and each molecule should be fully understood if the overall system is to be modeled properly.

Functional Choice One limitation, as mentioned prior, to any DFT work is the exchange-correlation functional of choice. With a set of molecules deeply inside the organic family, it becomes pertinent to model electrons effectively. While the PBE functional has been seen to do this, for future works a meta-GGA would be ideal to reflect a system with high levels of resonance. One such model that would be apt for this line of work is the SCAN functional.

Environment Modeling Because of the nature of Thyroid synthesis, these complex reactions do not occur in a vacuum. Solvents, polarity, and natural variation all serve to perturb the system away from ideal. In future work, a goal would be to capture the dynamics of these systems in hopes to address how alterations can proportionally change the overall reaction as well as its mechanistic steps.

Conclusion

This work employed molecular dynamics simulation and density functional theory calculations to investigate the structural and electronic properties of the IYD active site and FMN cofactor as they pertain to reductive dehalogenation. Together, the two approaches produced a consistent and complementary computational picture broadly supportive of the established mechanistic framework.

The MD simulation of oxidized IYD over a 1 ns production trajectory yielded a well-equilibrated system, confirmed by convergence of mean temperature to 299.87 K and system density to 0.9999 g/cm^3. Analysis of the minimum potential energy configuration extracted at t = 650 ps revealed that the key polar interactions necessary for FMNsq stabilization were preserved throughout the simulation. The substrate ammonium group was positioned 2.9 Å from the flavin O4, consistent with a strong electrostatic interaction, while the substrate carboxylate was located 3.2 and 4.1 Å from the flavin N3 and O4 positions respectively. The threonine side chain and arginine guanidinium group were also positioned proximal to their respective flavin coordination sites, collectively supporting the preservation of active site geometry required for transition-state stabilization.

The DFT analysis of T-FMN Hq demonstrated reliable convergence across the 600–700 eV cutoff range, with stable HOMO-LUMO gaps of approximately 1.565 ± 0.02 eV and decreasing residual forces at higher cutoff values. An anomalous result at 650 eV across multiple convergence metrics like energy, RMSD, and optimization steps is most consistent with entrapment in an alternative local minimum and warrants further investigation. Most significantly, electron density mapping revealed preferential accumulation of negative charge at the carbon-carbon bond adjacent to the isoalloxazine nitrogen atoms, consistent with the theorized radical transition state and directly supporting the sequential single-electron transfer mechanism.

Together, these results confirm that the IYD active site maintains its catalytically relevant geometry under simulation conditions and that the electron density distribution of FMN Hq is consistent with its proposed role as a single-electron donor. Future work should extend both approaches to the semiquinone and oxidized FMN states, incorporate active site residues and substrate into the quantum chemical model, and apply higher-level functionals to capture the full electronic complexity of the system.

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